Junie: Your AI Coding Agent in JetBrains IDEs

Autonomous Coding with Safety Controls

Transform Your Development Workflow
with AI-Powered Automation

Four-Hour Hands-On Workshop

🎯 What You'll Master Today

Core Concepts

- 🔍 **Ask Mode**: Analysis & insights
- ⚡ **Code Mode**: Automated changes
- 📋 **Guidelines**: Team conventions
- 🛡️ **Safety**: Approvals & controls

Hands-On Labs

- ☕ **Java**: Spring Boot APIs
- 🐍 **Python**: Refactoring & testing
- ⚛️ **React**: Forms & validation
- 🔧 **MCP**: External tool integration

📚 Prerequisites Check

✅ **JetBrains IDE** (IntelliJ IDEA, PyCharm, WebStorm, GoLand, PhpStorm, RustRover, RubyMine)

✅ **JetBrains AI** subscription (AI Credits model — 1 Credit = $1 USD)

✅ **Development Environment** (Java 17+ / Python 3.8+ / Node 18+)

🆓 AI Free: 3 AI Credits/month — or use BYOK (Bring Your Own Key) with any provider

💡 Quick Setup: We'll install Junie together in the first 5 minutes

🚀 Getting Started — Two Paths

IDE Plugin

1. AI Chat panel → Agent dropdown → Select **Junie**
2. Or: Settings → Plugins → Search "Junie"
3. Start prompting

Junie CLI (Beta)

1. `brew tap jetbrains/junie && brew install junie`
2. Or: `curl -fsSL https://junie.jetbrains.com/install.sh | bash`
3. Run `junie` in any project

Available Models (LLM-agnostic):

GPT / GPT Codex | Opus / Sonnet | Gemini Pro / Flash | Grok

Aliases auto-update to latest versions

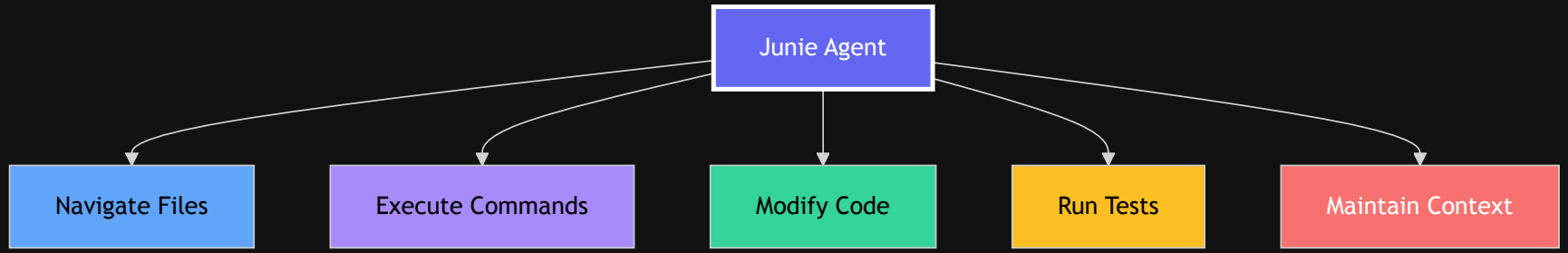
Today’s Journey

Four-Hour Workshop Timeline

☕ Break after Core Workflow segment

🤖 What is Junie?

JetBrains' AI-Powered Coding Agent



🆕 What's New in Junie (March 2026)

Junie CLI (Beta)

- Standalone terminal agent
- LLM-agnostic — BYOK support
- Plan mode, brave mode, prompt history
- One-click migration from Claude Code

AGENTS.md

- New open-standard guidelines format
- Replaces `.junie/guidelines.md` (legacy)
- Shared across AI coding agents
- Auto-import from other agents

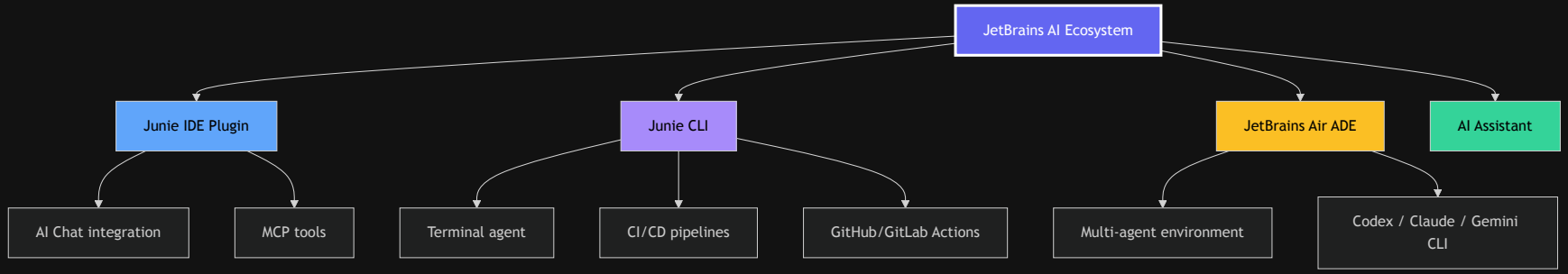
Agent Skills & Slash Commands

- Reusable task-oriented extensions
- `.junie/skills/{name}/SKILL.md`
- Custom slash commands in CLI
- Project-level and user-level scopes

CI/CD & Beyond

- GitHub Action (`@junie-agent`)
- GitLab CI (`#junie` in comments)
- Junie merged into AI Chat (Dec 2025)
- AI Credits pricing model

🗺️ JetBrains AI Landscape (2026)



Air = agentic dev environment (macOS public preview, March 2026) built on Fleet codebase, using Agent Client Protocol (ACP)

Air: Agentic Development Environment

Multi-Agent Sessions

- Work with Claude, Codex, Gemini, Junie
- Multiple agent sessions side by side
- Each in its own workspace
- You choose which agent fits each task

Isolated Workspaces

- **Local:** Direct project access (fastest)
- **Git Worktree:** Isolated branch
- **Docker:** Full container isolation
- Agents work without conflicts

Permission Modes

- Ask Permission / Auto-Edit
- Plan (read-only analysis)
- Full Access (autonomous)
- Similar to Junie's safety controls

Built-In Extras

- Web preview for frontend work
- Inline diff review + commenting
- MCP server integration
- @-mention files, branches, symbols

macOS public preview — air.dev

🔄 Two Operating Modes

🔍 Ask Mode

Read-Only Analysis

- Explain complex code
- Analyze architecture
- Find bugs & issues
- Review security
- Check test coverage

Perfect for understanding before changing

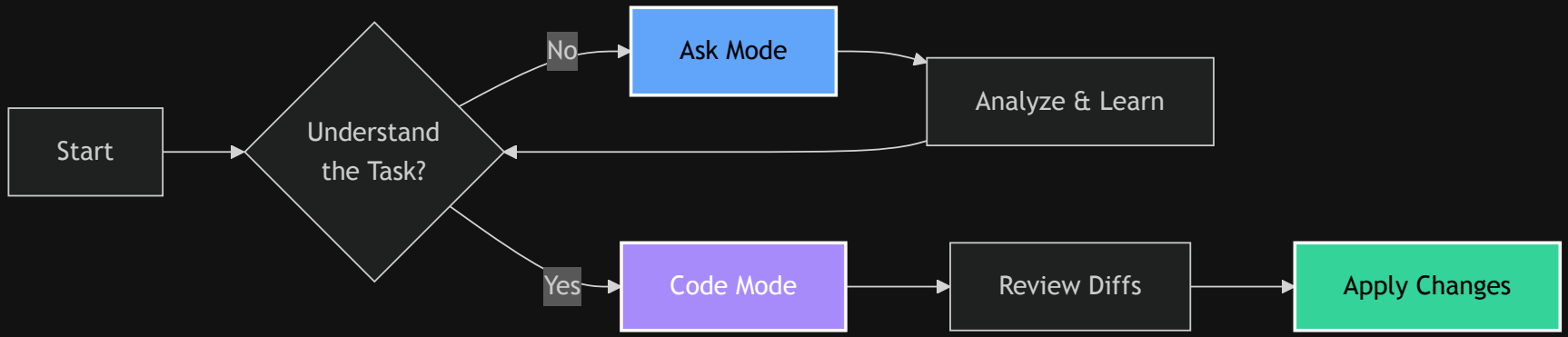
⚡ Code Mode

Make Changes

- Implement features
- Fix bugs
- Refactor code
- Generate tests
- Update dependencies

Executes multi-step plans with diffs

💡 Pro Workflow



✨ Always start with Ask mode for complex tasks

📝 Project Guidelines

New: `.junie/AGENTS.md` (open standard) | **Legacy:** `.junie/guidelines.md` (still supported)

```
## Technology Stack
- Framework: Spring Boot 3.2
- Testing: JUnit 5 + AssertJ

## Conventions
- REST: /api/v1/{resource}
- DTOs: Java records
- Tests: Given-When-Then
```

Lookup order: `.junie/AGENTS.md` → `AGENTS.md` (project root) → `.junie/guidelines.md` (legacy)

❌ Without Guidelines

Inconsistent patterns

Multiple review cycles

Style conflicts

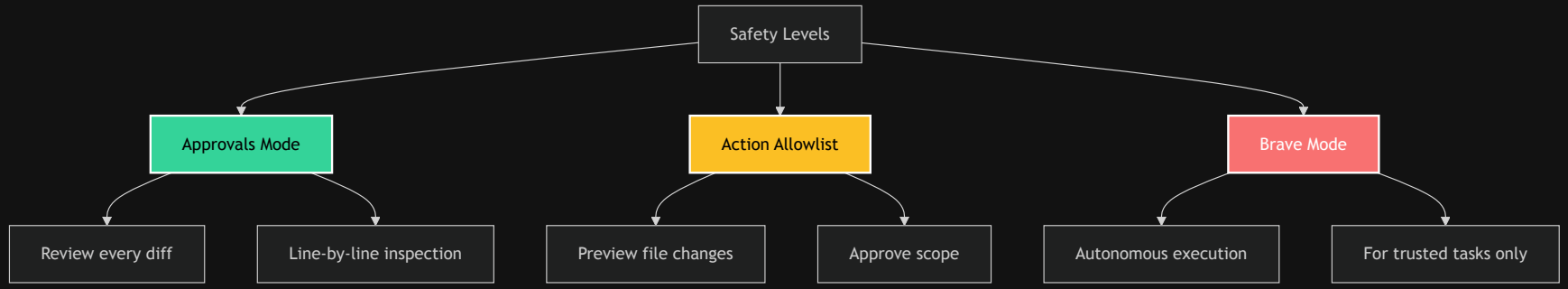
✅ With Guidelines

Consistent output

70% fewer reviews

Team alignment

Safety Controls



⚡ When to Use Each Mode

Mode	Good For	Not For
Approvals	• Critical business logic • First-time tasks • Learning Junie	• Repetitive formatting • Simple refactors
Allowlist	• Known file sets • Defined scope • Team reviews	• Exploratory changes • Unknown impact
Brave	• Test generation • Formatting • Documentation	• Production code • Database changes • First attempts

🔌 MCP: Model Context Protocol

Available Tools

📚 context7

- Real-time library docs
- 1000+ libraries
- Version compatibility
- Optional API key (higher rate limits)

🎭 Playwright

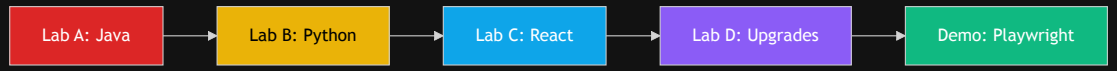
- E2E test generation
- Browser automation
- Accessibility-first
- TypeScript output

Configuration

```
{
  "mcpServers": {
    "context7": {
      "command": "npx",
      "args": ["-y", "@upstash/context7-mcp"]
    },
    "playwright": {
      "command": "npx",
      "args": ["@playwright/mcp-server"]
    }
  }
}
```

Settings → Junie → MCP → Edit mcp.json

🧪 Lab Overview



Timing:

- Lab A: 45-60 min
- Lab B: 30-45 min
- Lab C: 30-45 min
- Lab D: 20-30 min
- Demo: 20-30 min

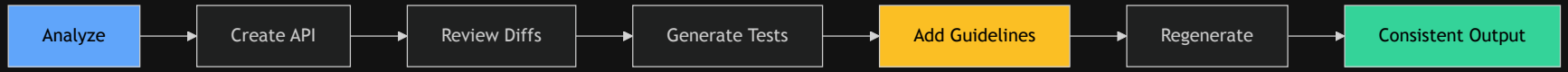
☕ Java
Spring Boot
REST APIs
JUnit + AssertJ

🐍 Python
PEP 8
Type hints
pytest

⚛ React
TypeScript
Forms
Testing Library

🔧 MCP
context7
Upgrades
Migration

☕ Lab A: Spring Boot Journey



Without Guidelines

- Inconsistent patterns
- Various styles
- More review cycles

With Guidelines

- Consistent code
- Team standards
- Faster approval

🐍 Lab B: Python Transformation

Before 😱

```
def calc(x,y,op):
    if op=="add": return x+y
    elif op=="sub": return x-y
```

- No type hints
- Poor naming
- No tests
- No docs

After ✨

```
def calculate(
    x: float,
    y: float,
    operation: str
) -> Optional[float]:
    """Calculate result.

    Args:
        x: First number
        y: Second number
        operation: Operation

    Returns:
        Calculation result
    """
```

Lab C: React Forms

```
interface RegistrationForm {
  email: string;      // valid email
  password: string;   // min 8, special char
  confirm: string;    // must match
  terms: boolean;     // required
}
```

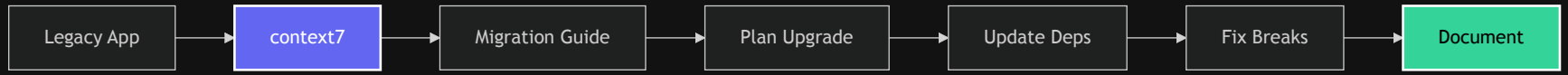
React Hook Form
Form state

Zod
Validation

Testing Library
90% coverage

Accessibility
ARIA compliant

🔧 Lab D: Smart Upgrades with context7



React 17 → React 18

Legacy → Modern

🎭 Demo: Playwright Magic

Generate E2E Tests in 5 Minutes

```
test('successful login', async ({ page }) => {
  const loginPage = new LoginPage(page);
  await loginPage.goto();
  await loginPage.login('user@example.com', 'pass');

  await expect(page).toHaveURL('/dashboard');
  await expect(page.getByRole('heading', { name: 'Welcome' }))
    .toBeVisible();
});
```

✅ Page Objects | ✅ Accessibility | ✅ TypeScript | ✅ Retry Logic

💡 MCP vs CLI: IDE plugin needs MCP to reach the browser. CLI agents (Junie CLI, Claude Code, Codex) can run `npx playwright test` directly — full test runner, cheaper on tokens. Use MCP for interactive browser exploration; use CLI for running tests.

⚔️ Junie's Differentiators

Feature	Junie	Claude Code	Copilot Agent	Cursor
IDE + CLI + CI/CD	✅ **All three**	CLI only	IDE + CI	IDE only
LLM-agnostic	✅ **BYOK**	Claude native, others via routing	GPT native, others via routing	✅
Test execution	✅ **Yes**	✅ **Yes**	✅ **Yes**	✅ **Yes**
MCP tools	✅ **Yes**	✅ **Yes**	✅ **Yes**	✅ **Yes**
JetBrains native	✅ **Full**	via ACP	Plugin	❌

🎯 Common Patterns

TDD Flow

1. Ask: "What tests?"
2. Code: "Generate tests"
3. Code: "Implement"

Red → Green → Refactor

Safe Refactoring

1. Generate tests
2. Identify issues
3. Refactor safely

Tests stay green

Upgrades

1. context7 check
2. Update deps
3. Fix breaks

Systematic migration

🚀 CI/CD Integration — Junie GitHub Action

```yaml
name: Junie Agent
on:
  issue_comment:
    types: [created]
  pull_request_review_comment:
    types: [created]

jobs:
  junie:
    if: contains(github.event.comment.body, '@junie-agent')
    runs-on: ubuntu-latest
    steps:
      - uses: JetBrains/junie-github-action@v1
        with:
          junie-api-key: ${{ secrets.JUNIE_API_KEY }}
```

GitHub
`@junie-agent` in PRs/issues

GitLab
`#junie` in MR comments

CLI Headless
`junie --auth="$KEY" "prompt"`

✨ Key Takeaways

✅ **Ask before Code** - Understand first, implement second

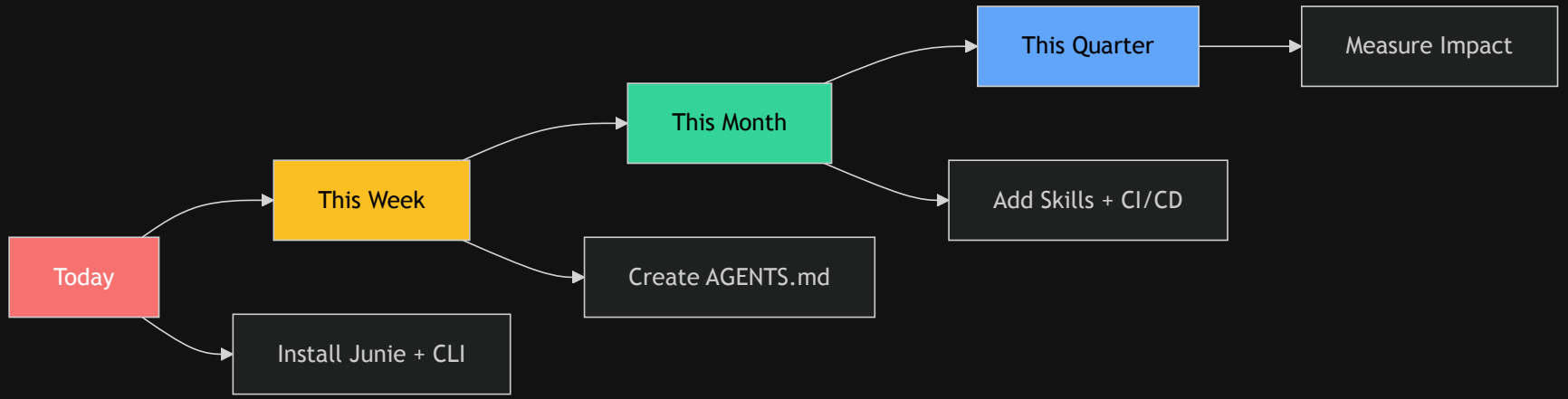
✅ **Guidelines drive consistency** - Encode your team's DNA

✅ **Build trust gradually** - Approvals → Allowlist → Brave

✅ **MCP extends capabilities** - Leverage external tools

✅ **Safety first** - Review diffs, maintain control

🎯 Your Next Steps



🚀 Start with Approvals Mode → Build Trust → Scale Up → Automate with CI/CD

💬 Discussion Time

Let’s Explore Together:

🤔 IDE plugin vs CLI vs CI/CD — which fits your workflow?

🛡️ Where do you always want human review?

📋 How could AGENTS.md guidelines help your team?

🔧 Which MCP tools and Agent Skills would you create?

🚀 How will multi-agent environments (Air, ACP) change development?

📚 Resources Hub

Documentation

- 📖 Junie Docs
- 📝 Guidelines & Memory
- 🔧 Agent Skills
- 🖥️ CLI Quickstart

CI/CD & Air

- 🐙 GitHub Action
- 🦊 GitLab CI
- 🌐 Air Getting Started

GitHub

- 🗂️ Guidelines Catalog
- 📚 Context7 MCP
- 🧪 This repository

Community

- 💬 JetBrains AI Slack
- 🏷️ Stack Overflow: `jetbrains-junie`
- 📧 Support team

🙏 Thank You!

AI amplifies good practices

💡 Remember:
Invest in your guidelines today,
reap consistency forever!

Questions? Let's explore together! 🚀

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AI agents aren't here to replace developers—
they're here to make us **better** developers.

🎁 Bonus: Quick Reference

IDE Shortcuts

- `Ctrl+Alt+J` - Open Junie
- `Ctrl+Enter` - Execute
- `Esc` - Cancel operation ### CLI Shortcuts • `Shift+Tab` - Plan mode
- `Ctrl+B` - Brave mode
- `Ctrl+R` - Prompt history
- `Ctrl+T` - Session transcript

Common Commands

"Analyze this file"
"Add tests with 90% coverage"
"Refactor to best practices"
"Fix all type errors"

MCP Tools

```
{
  "context7": "Library docs",
  "playwright": "E2E tests",
  "search": "Web search",
  "custom": "Your tools"
}
```

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